



बिहार पुलिस भर्ती



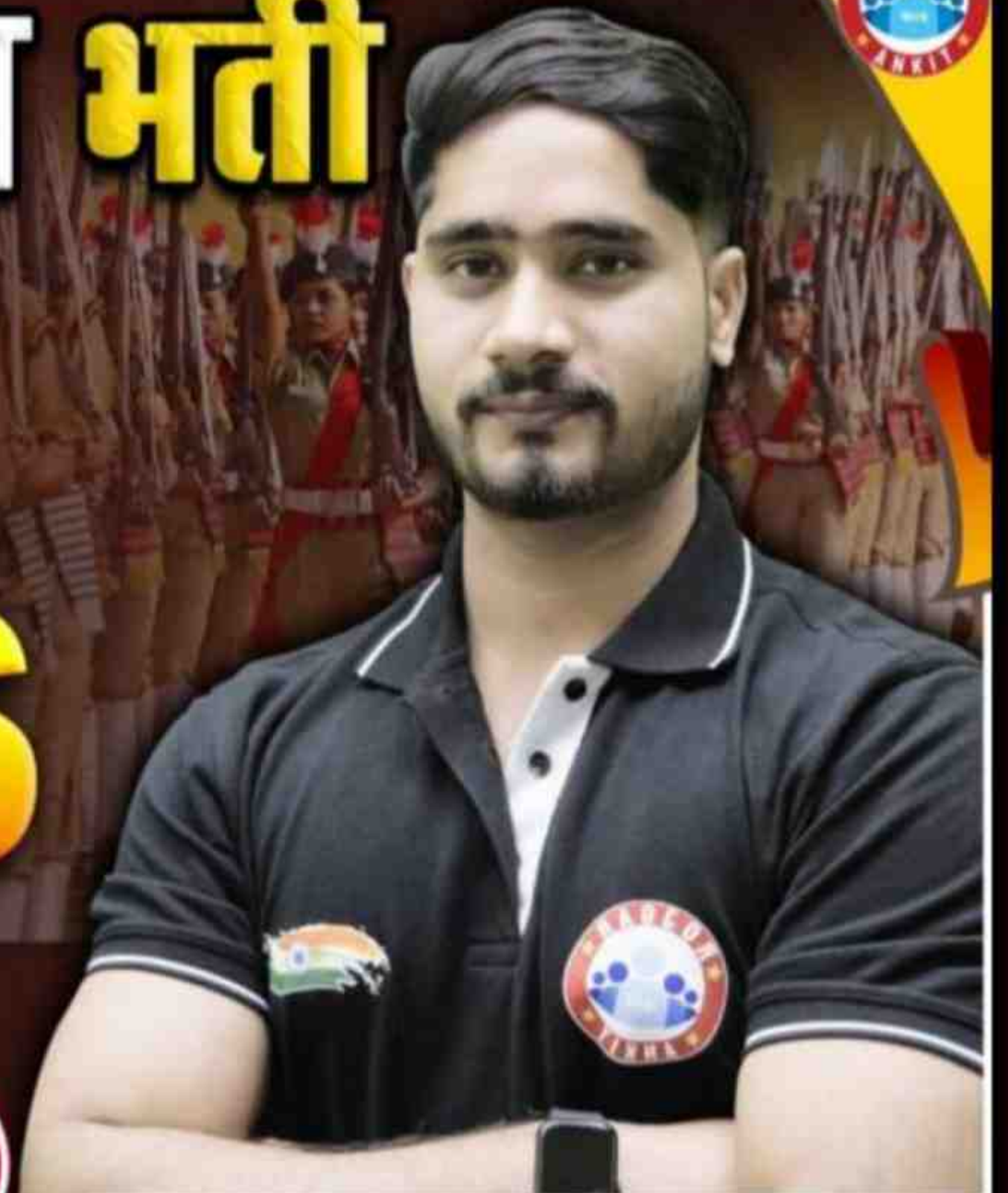
बुद्धा बैच

PHYSICS

Heat & Temperature

ऊष्मा एवं तापमान

02



LIVE 26-08-2023 6PM

Heat & Temp.

उष्मा एवं तापमान

1. 0°C + 273 = 273K ✓

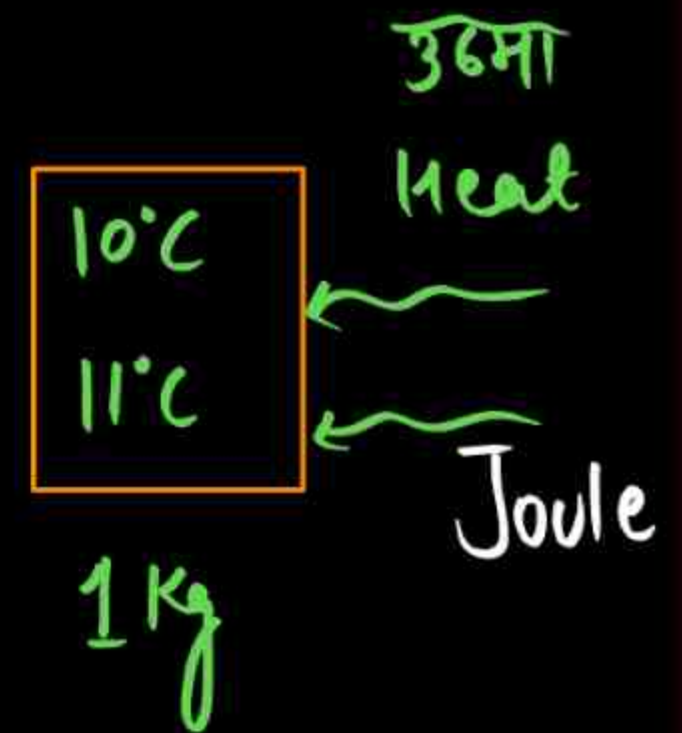
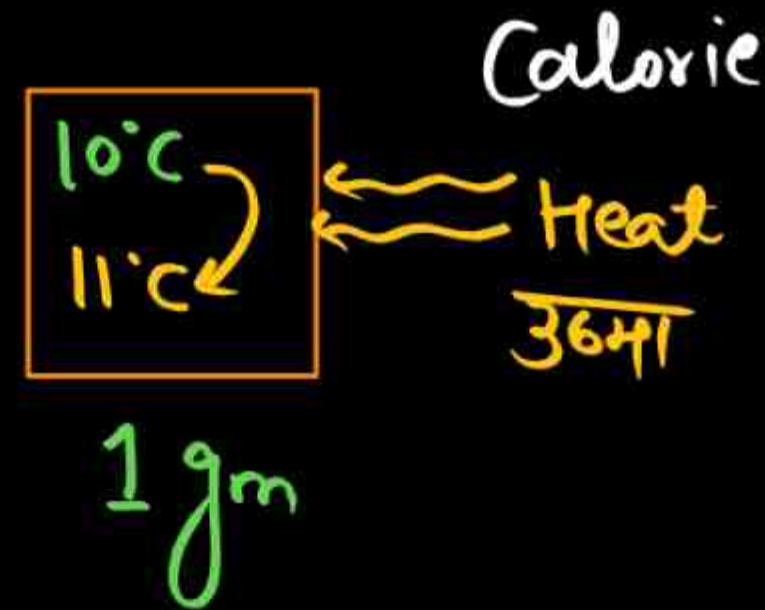
2. परमशून्य ताप = 0K
Absolute Temp. -273°C

विशिष्ट ऊष्मा (Specific Heat)

- किसी 1 gm की वस्तु का ताप 1°C बढ़ने के लिए आवश्यक ऊष्मा की मात्रा (Amount of heat required to increase temperature of 1 gm body by 1°C .)

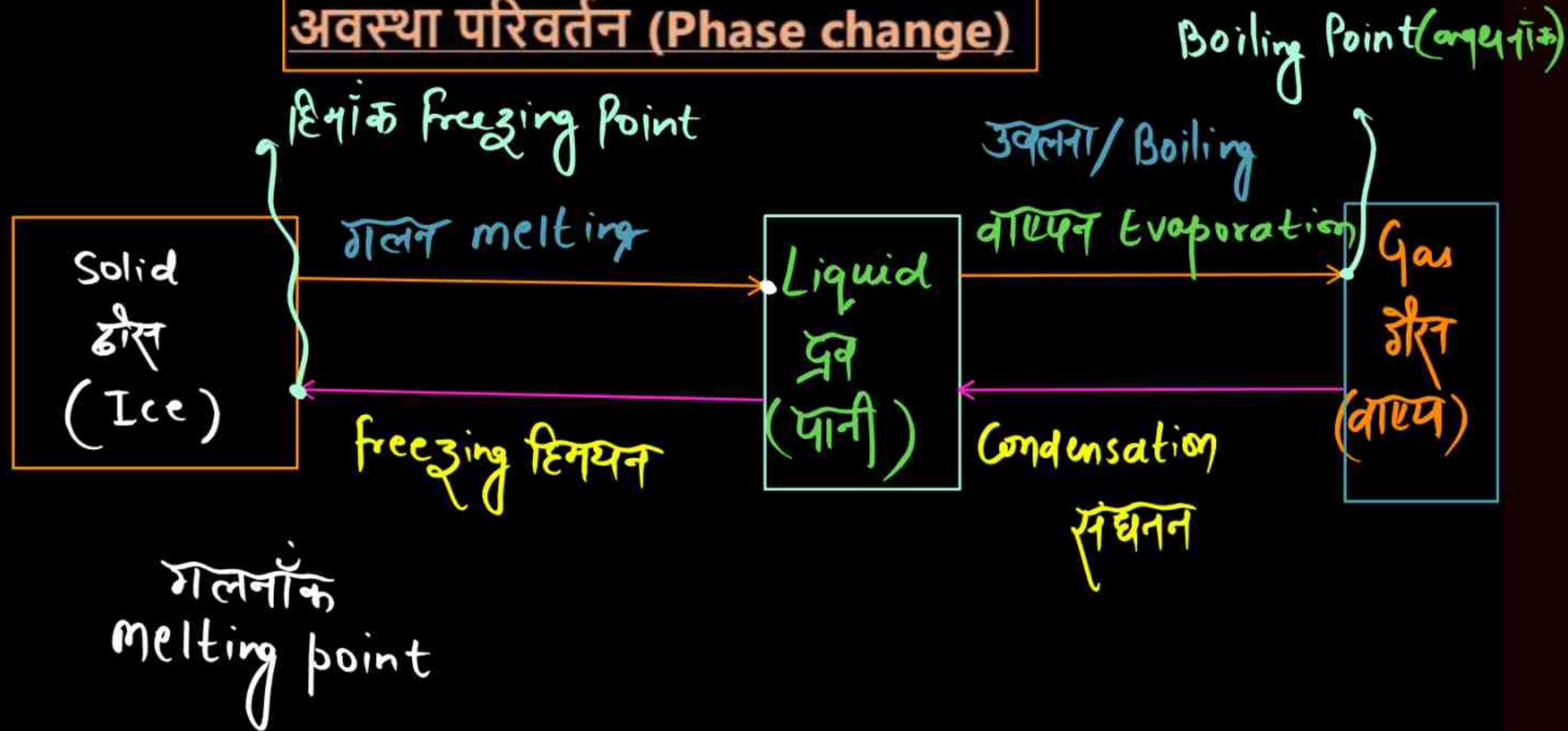
- S.I Unit =

- $C_{\text{water}} = 4180 \frac{\text{J}}{\text{kg-K}}$

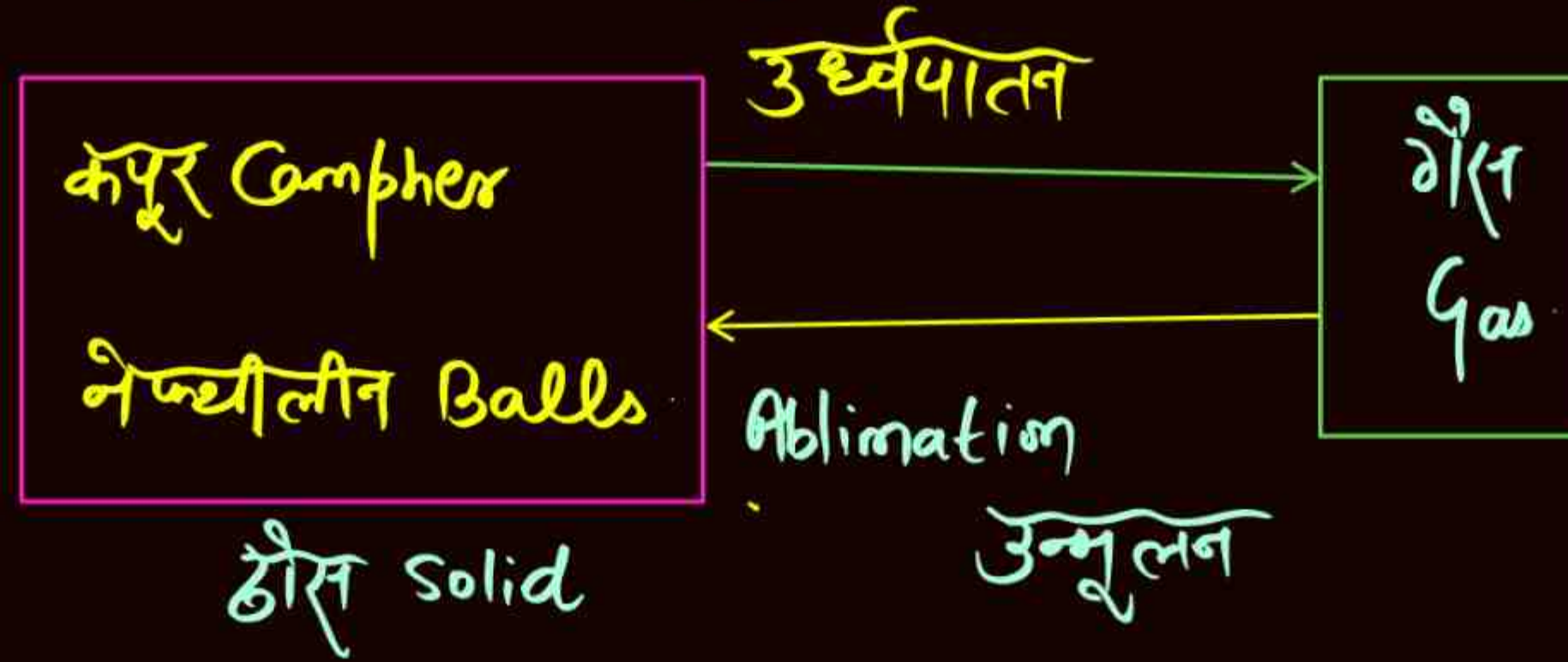


BIHAR POLICE | PHYSICS

अवस्था परिवर्तन (Phase change)



★ Sublimation



Ex:- dry Ice शुष्क बर्फ

BIHAR POLICE | PHYSICS

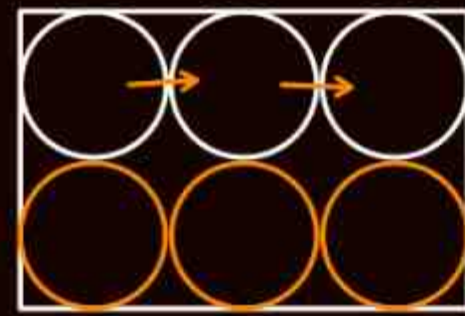
Note:-

Pressure दाब $\uparrow$ ----- $\rightarrow$ **B.P** $\uparrow$ कथनांक

दाब $\uparrow$ ----- $\rightarrow$ **F.P** $\downarrow$ हिमांक

120°C
 $\uparrow$
100°C

 $\downarrow$
0°C
-1°C



Solid ટોફ
(બર્ફ) 0°C

Heat



Liquid ટોફ
(પાની) 0°C

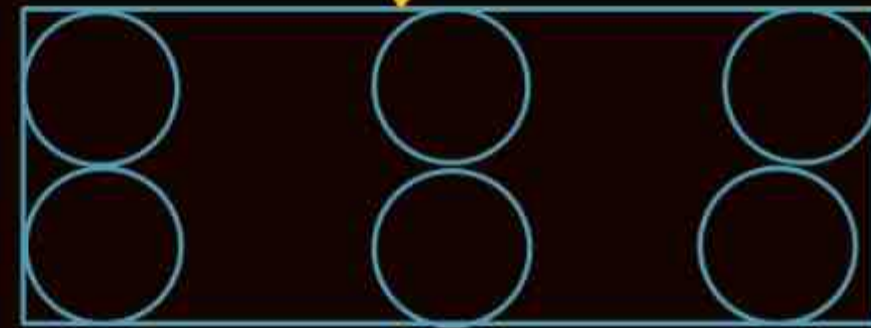
અવસ્થા પરિવર્તન ✓

Phase change

તાપમાન ↑
Temp. ↑ X

અવસ્થા પરિવર્તન

Phase change X



(10°C) પાની
liquid ટોફ

તાપમાન ↑
Temp. ↑ ✓

→ સંવેદનશીલ ટોફ
Sensible Heat

ગુપ્ત ટોફ
Latent Heat

संवेदनशील ऊष्मा (sensible)

Ex. 0°C (Ice) $\rightarrow$ 20°C (Water)

- ताप बदलेगा (Temp. change)
- अवस्था नहीं बदलेगी (Phase does not change)

